

Interactive TOPSIS Decision Analysis System

Objective

This project implements the TOPSIS multi-criteria decision-making method in Python.

The system allows users to:

- define custom criteria
- assign weights
- specify benefit/cost criteria
- evaluate alternatives
- generate Excel reports and visualizations

Project Structure

- Part 1: Interactive Excel input generation
- Part 2: TOPSIS ranking and visualization

Technologies

- Python
- Pandas
- NumPy
- Matplotlib
- OpenPyXL

Outputs

The system generates:

- ranked alternatives
- normalized matrices
- weighted matrices
- Excel reports
- charts

Methodology

TOPSIS (Technique for Order Preference by Similarity to Ideal Solution) ranks alternatives based on their distance from ideal and negative-ideal solutions.

References

- Hwang, C. L., & Yoon, K. (1981).
Multiple Attribute Decision Making: Methods and Applications.
Springer-Verlag.
- TOPSIS Method: <https://en.wikipedia.org/wiki/TOPSIS>
- Python Documentation: <https://www.python.org/>
- Pandas Documentation: <https://pandas.pydata.org/>
- NumPy Documentation: <https://numpy.org/>
- Matplotlib Documentation: <https://matplotlib.org/>